VASTRA
KOMPRESSOR AB - SWEDEN

Oil-injected rotary screw compressor, 110 kW, variable speed

# VRS-110

## Service Manual

*Translation of the original instructions (Swedish: Servicehandbok)*

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| Valid for | **VRS-110, serial no. from VRS110-220001** |
| Controller | **Vastra Vcon 4, firmware 5.0** |
| Drive | **Vastra VD 110 variable speed drive, integrated** |
| Replaces | **Preliminary edition, 2022-04** |

# Contents


**1 Safety 3**
1.1 Safety symbols 3
1.2 Signal words 3
1.3 General safety precautions 3
1.4 Before maintenance and repair 4
1.5 Personnel 4

**2 General description 5**
2.1 Introduction 5
2.2 Air flow and oil system 5
2.3 Sensors and switches 5
2.4 Technical data 5

**3 Installation requirements 6**
3.1 Compressor room and ventilation 6
3.2 Electrical supply 6
3.3 Cable routing 6

**4 Vcon 4 controller 7**
4.1 Control panel and menus 7
4.2 Pressure regulation 7
4.3 Dew point protection 7
4.4 Event log, service port and firmware 7

**5 Maintenance 8**
5.1 Maintenance schedule 8
5.2 Oil 8
5.3 Changing the oil and the oil filter 9
5.4 Replacing the air filter element 9
5.5 Replacing the separator element 9
5.6 Cleaning the coolers 9
5.7 Safety valve, minimum pressure valve and oil stop valve 9
5.8 Replacing the discharge temperature sensor TS-2 10

**6 Fault finding 10**
6.1 Fault and warning codes 10
6.2 Before fault finding 11
6.3 Fault finding procedures 11

**7 Variable speed drive 12**
7.1 Description 12
7.2 Safety at the drive 12
7.3 Qualification and authorisation for work on the drive 13
7.4 Drive fault codes 13
7.5 E-31 VFD overcurrent and E-36 Earth fault 13
7.6 Maintenance of the drive cabinet 14

**8 Spare parts 14**
8.1 Ordering 14
8.2 Service parts 14
8.3 Recommended stock 14

**9 Warranty and documentation 15**
9.1 Warranty 15
9.2 Revision history 15
9.3 Contact 15

# 1 Safety

## 1.1 Safety symbols

The symbols below are used in this manual and on the labels on the compressor. Damaged or missing labels shall be replaced without delay. Label sets are ordered from Vastra service; state the serial number of the compressor.

General danger. Read the text that belongs to the symbol.

Dangerous electrical voltage. Electrical cabinet, motor terminal box and drive cabinet (chapter 7).

Hot surface. Element outlet, discharge pipe, separator vessel and oil cooler can reach more than 110 C.

Rotating parts. Cooling fan and coupling. The fan can also run when the compressor is stopped.

System under pressure. The separator vessel stays pressurised after stop until it is vented.

Automatic start. The compressor can start without warning in automatic mode and after a power failure.

Read the manual before you start the work.

Wear hearing protection when you work at a running compressor with the doors open.

Wear safety glasses and protective gloves when you handle hot oil and when you vent the compressor.

Open fire and smoking are not allowed near the oil system and the separator vessel.

## 1.2 Signal words

**DANGER**
Indicates a hazardous situation that, if not avoided, will result in death or serious injury.

**WARNING**
Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

**CAUTION**
Indicates a situation that, if not avoided, could result in minor injury or damage to the compressor.

**NOTE**
Gives information that is important for correct maintenance. Observe the note.

## 1.3 General safety precautions

- The operating company is responsible for that the compressor is operated and maintained only by competent persons who have read this manual.
- It is not allowed to make changes on the compressor or its control system without written approval from Vastra Kompressor AB.
- The compressed air is not suitable for breathing unless it is purified according to EN 12021.
- Never run the compressor above the maximum working pressure on the data plate, and never block, adjust or remove the safety valve.

- Never run the compressor with the doors or panels open, except for the short checks described in this manual.
- Do not use the emergency stop for normal stopping. The emergency stop does not vent the separator vessel.
- The variable speed drive contains capacitors that keep a dangerous voltage after the supply is switched off. See 7.2 before any work in the drive cabinet.

## 1.4 Before maintenance and repair

Carry out the following steps before every maintenance or repair task, if the task description does not say otherwise.

1. Stop the compressor with the stop key (O). Wait until the unloaded run-down is finished and the motor has stopped.
2. Switch off the main isolator and lock it in the off position. Hang a sign "Work in progress - do not start" on the isolator.
3. Close the air outlet valve to the network. Open the manual vent valve on the separator vessel and control on the display (PT-1) that the internal pressure is 0 bar.
4. Let the compressor cool down. The element and the oil stay hot for a long time after stop.

**WARNING**

Observe that the stop key alone does not make the compressor safe. In automatic mode, and after a power failure with auto restart activated, the compressor starts without warning. Always lock the main isolator.

## 1.5 Personnel

Maintenance and fault finding as described in chapters 5 and 6 may be carried out by service technicians who are trained on Vastra VRS compressors. Work in the electrical cabinet requires an electrically qualified person. Work in the drive cabinet, and the reset of the drive faults that cannot be reset from the control panel, requires in addition an authorisation for work on the drive, see 7.3.

# 2 General description



## 2.1 Introduction

The VRS-110 is an oil-injected, single-stage rotary screw compressor with variable speed. The compression element is driven by a 110 kW electric motor through a flexible coupling. The motor speed is controlled by the integrated Vastra VD 110 variable speed drive (VSD), so that the delivered air follows the air demand at constant net pressure. The compressor is air-cooled and is delivered as a complete package in a sound-insulated enclosure, with the Vastra Vcon 4 controller and the drive in a separate, lockable drive cabinet on the right side of the package.

## 2.2 Air flow and oil system

The air and oil flow is the same as on the fixed-speed VRS compressors. Air is drawn in through the air filter and the inlet valve into the element. The air and oil mixture leaves the element through the discharge pipe with the discharge temperature sensor TS-2, and is separated in the separator vessel and the separator element. The air passes the minimum pressure valve and the aftercooler to the air outlet. The oil returns through the thermostatic valve, the oil cooler and the oil filter to the element.

Because the element speed varies from 25% to 100%, the oil injection is controlled by the internal pressure and by an oil stop valve that closes when the motor stops. At low speed the discharge temperature is lower and the oil can stay below the dew point of the air for a long time; the controller then raises the minimum speed to keep the oil warm (function "dew point protection", see 4.3).

## 2.3 Sensors and switches

| Designation | Function | Location |
|---|---|---|
| PT-1 | Internal pressure | Separator vessel, top cover |
| PT-2 | Net pressure (regulation) | Air outlet, after the minimum pressure valve |
| TS-1 | Air inlet temperature (ambient) | Air filter housing |
| TS-2 | Discharge temperature (element outlet) | Discharge pipe, approx. 150 mm after the element outlet flange, in a thermowell |
| TS-5 | Drive cabinet temperature | Drive cabinet, upper part |
| PS-3 | Oil filter differential pressure switch | Oil filter head |
| PS-4 | Air filter vacuum switch | Air filter housing, outlet side |
| B21 | Motor PTC thermistors | Motor winding, connected to the drive |

## 2.4 Technical data

| Parameter | Value |
|---|---|
| Working pressure range | 4 - 13 bar(e), set by the user |
| Free air delivery at 7 bar (ISO 1217, Annex E) | 5.6 - 19.8 m3/min |
| Free air delivery at 10 bar | 4.9 - 16.4 m3/min |
| Free air delivery at 13 bar | 4.2 - 13.3 m3/min |
| Motor | 110 kW, IE3, 2-pole, IP55, for operation with variable speed drive |

| Parameter | Value |
|---|---|
| Motor speed range | 900 - 3,600 rpm |
| Supply | 400 V 50 Hz or 460 V 60 Hz, 3-phase, TN system |
| Max. input current | 196 A at 400 V / 171 A at 460 V |
| Recommended main fuses (gG) | 250 A at 400 V / 225 A at 460 V |
| DC bus voltage in operation | Approx. 540 V DC (400 V supply), approx. 620 V DC (460 V supply) |
| Oil capacity (system) | 62 l |
| Approved oil | Vastra SynLube 46 only |
| Discharge temperature, normal operation | 75 - 95 C (pre-warning E-13 at 105 C, shutdown E-14 at 110 C) |
| Permitted ambient temperature | +1 to +46 C |
| Cooling air flow | 10.5 m3/s |
| Sound pressure level (ISO 2151) at full speed | 73 dB(A) |
| Air outlet connection | DN 80 flange |
| Dimensions L x W x H | 2,400 x 1,300 x 1,950 mm |
| Weight | 2,280 kg |



# 3 Installation requirements

The installation is described in the Installation Instructions 9821 0110 03. The points below are often the cause of service calls and shall be controlled at every service visit.

## 3.1 Compressor room and ventilation

The compressor room shall be dry, free from dust and frost-free. The cooling air inlet and outlet shall have a free area of at least 1.6 m2 each. With an outlet duct, the duct shall have a separate fan with a capacity of at least 10.5 m3/s, and the pressure drop in the duct shall not exceed 30 Pa. The hot cooling air shall not be recirculated to the inlet. Most E-13 and E-14 events in summer are caused by recirculation or by a duct fan that is not running.

The temperature in the drive cabinet depends on the room temperature. Above 40 C in the room, the drive reduces the maximum speed (derating) and warning E-34 can be shown.

## 3.2 Electrical supply

The supply cable and the main fuses shall be sized according to 2.4 and the local regulations. The drive is designed for TN supply systems. For IT systems the EMC filter must be disconnected by Vastra service before connection. Power factor correction capacitors on the same supply must have detuning reactors; otherwise they can resonate with the drive and give E-33.

## 3.3 Cable routing

Motor cables and supply cables shall be routed separately from signal and sensor cables. Do not lay sensor cables in the same cable tray as the motor cable, and keep them away from hot surfaces such as the discharge pipe and exhaust air ducts. Use the original cable clips. A damaged sensor cable must be replaced, not repaired with tape.

# 4 Vcon 4 controller

## 4.1 Control panel and menus

The Vcon 4 on the VRS-110 has the same keys, LEDs and basic menu structure as the Vcon 3 on the fixed-speed VRS compressors: Measurements, Counters, Event log, Settings (customer code) and Service (service code). In addition, the main screen shows the motor speed in percent and the actual flow in m3/min, and the menu Measurements has a page "Drive" with the motor current, the DC bus voltage, the heat sink temperature and the drive cabinet temperature (TS-5).

## 4.2 Pressure regulation

The controller keeps the net pressure (PT-2) at the setpoint C1.10 by changing the motor speed. When the air demand is below the minimum flow, the compressor unloads and stops after the unloaded run time C1.03. The compressor restarts when the pressure falls 0.3 bar below the setpoint. The regulation (PI) is set at the factory and shall normally not be changed.



| Parameter | Description | Range | Factory |
|---|---|---|---|
| C1.10 | Net pressure setpoint | 4.0 - max. working pressure - 0.5 | 7.0 bar |
| C1.03 | Unloaded run time before stop | 10 - 300 s | 30 s |
| C1.05 | Auto restart after power failure | Off / On | Off |
| C1.06 | Restart delay after power return | 10 - 600 s | 30 s |
| C1.08 | Control mode | Local / Remote / Modbus | Local |
| C1.12 | Maximum speed limit | 60 - 100 % | 100 % |

## 4.3 Dew point protection

At low speed and low ambient temperature the oil temperature can stay so low that water from the air condenses in the oil. The controller calculates the dew point from TS-1 and the internal pressure, and raises the minimum speed when the discharge temperature is less than 10 C above the dew point. The function cannot be switched off.

## 4.4 Event log, service port and firmware

The event log stores the last 100 events with the date and time, the code, the running hours and the values of PT-1, PT-2, TS-1, TS-2 and the motor speed. Drive faults are also stored with the motor current and the DC bus voltage at the moment of the fault. The log can be exported with the VastraLink service software through the service port X10 on the controller board in the control compartment (not in the drive cabinet).

Controller firmware updates are made through the service port X10 with VastraLink and the Vastra service cable, which is available to authorised service partners. The compressor must be stopped during the update; the update takes approximately 20 minutes and the customer settings are kept. Vastra informs about new firmware versions and their content by service bulletins. Always read the bulletin before the update.

# 5 Maintenance

## 5.1 Maintenance schedule

The intervals are maximum intervals for normal operating conditions. The task is due at the interval in running hours or at the time interval, whichever comes first. The controller counts the running hours down for each task and shows warning E-19 when a task is due.

| Interval | Task | Part no. | See |
|---|---|---|---|
| Daily | Control the oil level in the sight glass (compressor stopped for 2 minutes). Control the display for warnings. | | 5.2 |
| 500 h / monthly | Clean the pre-filter mats in the enclosure air inlet and the filter mats of the drive cabinet. | | 5.6, 7.6 |
| 2,000 h | Replace the air filter element, or earlier at warning E-18. | 41-1512 | 5.4 |
| 2,000 h | Clean the oil cooler and the aftercooler. Test the electronic water drain. Control the drive cabinet fans. | | 5.6, 7.6 |
| 4,000 h / 12 months | Change the oil. | 41-0046 | 5.3 |
| 4,000 h / 12 months | Replace the oil filter, or earlier at warning E-07. | 41-1480 | 5.3 |
| 8,000 h | Replace the separator element. | 41-3075 | 5.5 |
| 8,000 h / yearly | Test the safety valve. Control the minimum pressure valve, the oil stop valve and the coupling element. | | 5.7 |
| 8,000 h / yearly | Control the sensors PT-1, PT-2 and TS-2 against reference instruments. | | 6.3.1 |
| Yearly | Control the tightness of the power connections in the drive cabinet (authorised person only). | | 7.3, 7.6 |
| 24,000 h | Overhaul of the compression element bearings and shaft seal (Vastra service only). | | |

In dusty or humid environment, and when the ambient temperature is continuously above 40 C, the intervals for the air filter, the coolers, the drive cabinet filter mats and the oil shall be shortened. Contact Vastra service for advice.

## 5.2 Oil

Only Vastra SynLube 46 (part no. 41-0046, 20 l can) is approved for the VRS-110. The high element speed at full load and the long periods at low speed demand a synthetic oil. Mineral oil is not allowed and makes the warranty invalid.

Control the oil level with the compressor stopped for 2 minutes. The level shall be in the upper half of the sight glass. An oil consumption of more than 0.5 l per 100 running hours is not normal; control the separator element and the scavenge line.

## 5.3 Changing the oil and the oil filter

1. Run the compressor at full speed until it is warm (discharge temperature above 70 C).
2. Stop, isolate and vent the compressor according to 1.4. Control that PT-1 shows 0 bar.
3. Open the oil filler plug slowly by one turn. Remaining pressure escapes through the vent hole in the plug thread.
4. Drain the oil from the separator vessel drain valve and from the oil cooler drain plug into a container.
5. Remove the oil filter 41-1480 with a strap wrench. Clean the sealing surface on the filter head.
6. Oil the gasket of the new filter. Screw it on by hand until the gasket touches, then a further 3/4 turn by hand.
7. Fill approximately 56 l of Vastra SynLube 46 until the level is in the upper half of the sight glass. Fit the filler plug with a new O-ring.
8. Run the compressor for 5 minutes, stop, wait 2 minutes and control the level. Top up if necessary.
9. Reset the service counters for oil and oil filter. Write the date and the running hours in the service record.



## 5.4 Replacing the air filter element

Stop and isolate the compressor. Open the cover clips, remove the cover and pull out the element 41-1512 with a turning movement. Clean the housing with a damp cloth, fit the new element and the cover, and reset the counter. Never clean the paper element with compressed air.

## 5.5 Replacing the separator element

The procedure is the same as for the VRS-75 and uses the same element 41-3075. Stop, isolate and vent the compressor (1.4); disconnect the scavenge line and the control line; remove the cover bolts (M16) and lift off the cover; replace the element and the gaskets; control the earthing staple; tighten the cover bolts crosswise to 110 N-m; clean the scavenge line restrictor (1.5 mm). After the start, control the oil flow in the scavenge line sight glass and note the separator dP from Measurements in the service record. A rising separator dP from one service to the next shows that the element comes to the end of its life.

## 5.6 Cleaning the coolers

Clean the oil cooler and the aftercooler from the fan side, against the normal direction of the cooling air, with compressed air at maximum 6 bar. Oily dirt is removed with a degreasing agent and water at low pressure; a high-pressure washer is not allowed. The cooling fan of the VRS-110 is speed-controlled; during cleaning it can start if the main isolator is not locked.

## 5.7 Safety valve, minimum pressure valve and oil stop valve

Test the safety valve once per year with the test lever, with the compressor running, or according to the national regulations if these are more strict. A valve that leaks after the test shall be replaced. Control that the oil stop valve closes when the motor stops: after a stop, no oil shall be pressed out of the inlet valve. An oil stop valve that does not close gives oil in the air filter housing.

### 5.8 Replacing the discharge temperature sensor TS-2

TS-2 (41-2207) is a PT1000 sensor that sits in a thermowell in the discharge pipe. It can be replaced without draining the oil.

1. Stop, isolate and vent the compressor (1.4). Let the discharge pipe cool down.
2. Disconnect the plug at the sensor. Unscrew the sensor from the thermowell. Do not unscrew the thermowell from the pipe.
3. Apply a small quantity of heat-conducting paste on the tip of the new sensor. Screw it in by hand, then a further 1/4 turn.
4. Connect the plug. Control that the cable is fixed with the original clips along its full route and does not touch hot parts.
5. Start the compressor and compare the reading with a handheld thermometer as described in 6.3.1.

# 6 Fault finding



### 6.1 Fault and warning codes

S = shutdown, the compressor stops. W = warning, the compressor continues to run. Codes E-30 to E-36 belong to the variable speed drive and are described in chapter 7. Codes that are not in the lists are not used on the VRS-110.

| Code | Type | Display text | Condition | Action / see |
|---|---|---|---|---|
| E-01 | S | Emergency stop | Emergency stop button pressed | Find out why it was pressed. Unlock the button, press Reset |
| E-02 | S | Supply fault | Phase missing or supply voltage outside +/-10% at the drive input | Control the supply and the main fuses |
| E-03 | S | Fan motor overload | Motor protection of the cooling fan | Control that the fan turns freely. Reset the motor protection |
| E-04 | W | Internal pressure sensor | PT-1 signal outside 4 - 20 mA | Control the cable and plug at X12. Replace PT-1 |
| E-05 | S | Main motor overload | Motor current above the limit for too long (drive I2t) | Control the element for heavy running. Control the supply voltage |
| E-06 | W | Net pressure sensor | PT-2 signal outside 4 - 20 mA | Compressor runs at fixed speed. Control cable X12, replace PT-2 |
| E-07 | W | Oil filter dP high | PS-3 closed, filter dP above 1.5 bar at operating temperature | Replace the oil filter (5.3) |
| E-09 | S | Internal pressure low | PT-1 below 2.5 bar 20 s after load | 6.3.3 |
| E-10 | W | Inlet temperature sensor | TS-1 open or short circuit | Dew point protection works with a fixed value. Control X13 |
| E-11 | S | Net pressure high | PT-2 above max. working pressure + 0.3 bar | 6.3.2 |
| E-12 | S | Low ambient temperature | TS-1 below +1 C. Start is blocked | Heat the compressor room |
| E-13 | W | Discharge temperature | TS-2 above 105 C (pre-warning) | 6.3.1 |
| E-14 | S | Discharge temperature high | TS-2 above 110 C | 6.3.1 |

| Code | Type | Display text | Condition | Action / see |
|---|---|---|---|---|
| E-17 | W | Drain alarm | Alarm contact of the electronic water drain | Clean the drain strainer. Control the drain valve |
| E-18 | W | Air filter restriction | PS-4 vacuum switch closed (65 mbar) | Replace the air filter element (5.4) |
| E-19 | W | Service due | A service interval counter has reached 0 | Carry out the task shown, reset the counter |
| E-20 | S | Motor temperature | PTC thermistors B21 in the motor winding | Clean the motor fan cover. Control the ambient temperature |
| E-21 | W | Too many starts | More than 6 motor starts in the last hour | Increase C1.03. Control the air demand and the receiver volume |
| E-22 | W | Remote communication | No remote signal or Modbus message for 10 s | Compressor runs in local mode. Control the remote connection |
| E-30 - E-36 | S/W | Drive faults | See 7.4 | 7.4, 7.5 |

## 6.2 Before fault finding

Read the event log before any action. Note the code, the time, the running hours, the motor speed and the temperatures at the moment of the event, and ask the operator what the compressor and the plant were doing (load, ambient temperature, power disturbances). Repeated events of the same code within a short time are more important than the single event and shall always be reported in the service report.

## 6.3 Fault finding procedures

### 6.3.1 E-13 and E-14 Discharge temperature

The controller stops the compressor when TS-2 measures more than 110 C at the element outlet (E-14). The pre-warning E-13 is shown at 105 C. On the VRS-110 a high discharge temperature appears first at full speed, which is when the cooling load is highest. Control in this order:

1. The ventilation of the compressor room and the ambient temperature (3.1). With ambient above 40 C the margin to E-14 at full speed is small. Control that hot cooling air is not recirculated and that the duct fan runs.
2. The coolers and the pre-filter mats. Clean them (5.6).
3. The oil level. Top up with Vastra SynLube 46 and look for oil leaks.
4. The cooling fan. The fan speed is controlled by the Vcon 4; control in Measurements that the fan speed reaches 100% when the discharge temperature is above 95 C.
5. The thermostatic valve. If the oil cooler stays cold while the discharge temperature is high, replace the insert.
6. The oil filter and the age of the oil.
7. The sensor TS-2. A TS-2 that reads too high gives E-14 although the compressor is not hot. With the compressor running loaded at full speed for at least 15 minutes, compare the TS-2 reading on the controller (Measurements > Discharge temperature) with a calibrated handheld contact thermometer at the discharge flange of the element. The handheld normally reads a few degrees lower, because it measures the flange surface. A difference above 8 C indicates a failing sensor: replace TS-2 (41-2207) according to 5.8.

After an E-14 shutdown, let the compressor cool for at least 15 minutes before restart. If E-14 comes back and no cause is found, export the event log and contact Vastra Technical Support. Do not raise the shutdown limit; it is fixed in the firmware.

### 6.3.2 E-11 Net pressure high

On the VRS-110 the pressure is regulated by the speed, and E-11 normally means that the compressor could not unload in time. Control the PT-2 reading against a reference gauge, the setpoint C1.10, the loading solenoid valve and the inlet valve. A sudden drop of the air demand to zero with a small receiver can give E-11 at full speed; the receiver volume shall be at least 1.5 m3 for the VRS-110.

### 6.3.3 E-09 Internal pressure low

The internal pressure does not reach 2.5 bar after loading. Control the inlet valve and its solenoid, the blow-off valve, the minimum pressure valve and the oil level. At start against an empty network the minimum pressure valve shall keep PT-1 above 4 bar; if it does not, replace the valve kit.

### 6.3.4 The pressure is unstable or the speed hunts

Control that PT-2 is not mounted after a filter or a dryer with a high pressure drop, that the receiver volume is sufficient, and that no other compressor on the network regulates in the same pressure band. The PI parameters shall only be changed after consultation with Vastra Technical Support.

### 6.3.5 The compressor does not start after a power failure

The compressor only starts by itself after a power failure if C1.05 is set to On; the restart delay C1.06 must pass first. A drive fault that occurred at the moment of the power failure must be reset before the compressor can start; see 7.4 for which faults can be reset from the control panel.



# 7 Variable speed drive

## 7.1 Description

The Vastra VD 110 drive is a frequency converter with a diode rectifier, a DC bus with capacitors, and an IGBT inverter that feeds the motor. A line choke and an EMC filter are built in. The drive is placed in the drive cabinet on the right side of the package, which has its own door with a key lock and its own cooling fans and filter mats. The drive communicates with the Vcon 4 controller over an internal bus (connector X14). All operation is done from the Vcon 4; the drive has no own keypad for the user.

## 7.2 Safety at the drive

**DANGER**

Dangerous voltage. The DC bus capacitors keep a dangerous voltage for several minutes after the supply is switched off. Before any work in the drive cabinet, switch off and lock the main isolator, wait at least the time given on the label on the drive cabinet door, and verify the absence of voltage on the DC bus with a suitable measuring instrument. The charge LED on the drive is not a safe indication that the capacitors are discharged.

**WARNING**

The motor terminals and the motor cable carry a dangerous voltage whenever the drive is energised, also when the motor stands still. Never open the motor terminal box with the main isolator switched on.

**CAUTION**

Do not carry out insulation resistance or high-voltage tests on the drive, and never with the motor cable connected to the drive. Disconnect the motor cable at the drive output before the motor insulation is tested.

## 7.3 Qualification and authorisation for work on the drive

Work in the drive cabinet may only be carried out by an electrically qualified person who is authorised by the operating company for work on variable speed drives according to the national regulations (for example EN 50110-1 in Europe or NFPA 70E in North America), and who is trained on the Vastra VD drive. This applies to the reset of the drive faults that cannot be reset from the control panel, to the control of power connections, and to the replacement of the drive cabinet fans.

Service technicians without this authorisation must not open the drive cabinet. They may read the event log and the values in Measurements > Drive, clean or replace the outer filter mats from outside, and report the fault.

## 7.4 Drive fault codes

| Code | Type | Display text | Condition | Reset |
|---|---|---|---|---|
| E-30 | S | Drive communication | No communication between Vcon 4 and drive for more than 2 s | Control panel |
| E-31 | S | VFD overcurrent | Output current above the peak current limit of the drive | Not possible from the control panel. See 7.5 |
| E-32 | S | DC bus undervoltage | Supply voltage dip or phase loss | Control panel. Auto restart if C1.05 = On |
| E-33 | S | DC bus overvoltage | DC bus voltage above the upper limit, often at supply overvoltage or resonance (3.2) | Control panel |
| E-34 | W | Drive cabinet temperature | TS-5 above 50 C; the drive reduces the maximum speed | Automatic |
| E-35 | S | VFD heatsink temperature | Heat sink temperature above the limit | Control panel, after cooling down |
| E-36 | S | Earth fault | Earth fault current in the motor circuit | Not possible from the control panel. See 7.5 |

For E-30, E-32, E-33 and E-35 the cause shall be removed before reset: control the connector X14 (E-30), the supply and the information from the site about grid events (E-32), the supply voltage and the power factor correction (E-33), the drive cabinet filter mats and fans and the room temperature (E-35). If the same code returns, do not reset again and again; contact Vastra Technical Support with the event log.

## 7.5 E-31 VFD overcurrent and E-36 Earth fault

The drive switches off the output immediately when the output current exceeds its peak current limit (E-31) or when it detects an earth fault current in the motor circuit (E-36). Possible causes of E-31 are a short circuit or an insulation fault in the motor or the motor cable, a blocked or heavy running element, a fault in the inverter, and supply disturbances (voltage dips) during operation at high speed. E-36 is caused by an insulation fault in the motor, the motor cable or the terminal box, often after humidity or condensate in the terminal box.

E-31 and E-36 cannot be acknowledged from the control panel or by remote reset. The Reset key has no effect while these codes are active. Reset requires access to the drive cabinet and must be carried out by an electrically qualified person authorised for work on the drive (see 7.3), after the cause has been investigated.

Before the authorised person arrives, the service technician or the operator can save time by noting from the event log the motor speed, the motor current and the DC bus voltage at the moment of the fault, by asking the site whether there was a power disturbance, and by controlling from outside that the element is not blocked (no unusual noise before the trip, no oil on the floor). Do not try to restart the compressor by switching the main isolator off and on; this does not reset the fault and stresses the drive.

**NOTE**

If E-31 or E-36 occurs repeatedly, the compressor shall not be reset again before Vastra Technical Support has been contacted. Send the event log export and the measured values.

### 7.6 Maintenance of the drive cabinet

The drive cabinet has two cooling fans and two filter mats (inlet and outlet). Clutter in the filter mats is the most common cause of E-34 and E-35. The outer filter mats can be cleaned or replaced from outside without opening the cabinet, every 500 h or monthly in dusty rooms. Once per year an authorised person (7.3) shall control the fans, the tightness of the power connections and the cabinet for dust and humidity. After a storage of more than one year without voltage, the DC bus capacitors must be reformed by Vastra service before the first start.

# 8 Spare parts

### 8.1 Ordering

Always state the type, the serial number (data plate inside the electrical cabinet door) and the part number. Use only genuine Vastra parts and Vastra SynLube 46. Spare parts for the drive are delivered only to authorised service partners. Parts that are not listed below are identified by serial number in Vastra PartsPortal.



### 8.2 Service parts

| Part no. | Description | Qty | Used in |
|---|---|---|---|
| 41-0046 | Vastra SynLube 46, synthetic compressor oil, 20 l can | 3 | 5.3 |
| 41-1480 | Oil filter, spin-on, VRS-110 | 1 | 5.3 |
| 41-1512 | Air filter element, VRS-110 | 1 | 5.4 |
| 41-3075 | Oil separator element with gaskets (common with VRS-75) | 1 | 5.5 |
| 41-2207 | Discharge temperature sensor TS-2, PT1000, with plug | 1 | 5.8, 6.3.1 |

### 8.3 Recommended stock

For a site with one or two VRS-110, Vastra recommends to keep on site one air filter element, one oil filter, 60 l of Vastra SynLube 46 and a set of drive cabinet filter mats. Sites where the compressors are critical for the production should also keep one discharge temperature sensor TS-2.

# 9 Warranty and documentation

## 9.1 Warranty

Vastra Kompressor AB gives a warranty for defects in material and manufacturing for 12 months from the date of commissioning, but at most 18 months from the date of delivery from the Vastra factory.

The warranty is valid on the following conditions:

- The commissioning report is sent to Vastra within 30 days from commissioning.
- Maintenance is carried out according to the maintenance schedule in this manual and is documented in the service record with date and running hours.
- Only genuine Vastra spare parts and Vastra approved oils are used.
- The compressor is installed and operated within the limits given in the technical data.

The warranty does not cover wear parts (filters, separator element, oil, coupling element), nor damage caused by wrong installation, insufficient ventilation of the compressor room, frost, condensate in the oil because of too low operating temperature, or the use of non-genuine parts. Warranty claims shall be sent by the authorised Vastra service partner within 30 days after the failure, with the serial number, the running hours and an export of the controller event log. Parts replaced under warranty are the property of Vastra and shall be kept for 60 days for a possible return.

For the VRS-110 the warranty also requires that the drive cabinet filter mats are maintained according to 5.1 and that work in the drive cabinet is done only by authorised persons (7.3).

## 9.2 Revision history

| Rev. | Date | Changes |
|---|---|---|
| - | 2022-04-04 | Preliminary edition, delivered with the first production units. |
| A | 2023-11-14 | First full edition. Chapter 7 completed: drive fault codes E-30 to E-36, reset restrictions for E-31 and E-36, qualification for work on the drive. 3: installation requirements added. 5.1: drive cabinet filter mats and yearly sensor control added to the schedule. 5.8 and 6.3.1: replacement and verification of TS-2 added. |

## 9.3 Contact

Vastra Kompressor AB, Technical Support, Verkstadsgatan 9, SE-721 36 Vasteras, Sweden.

Technical Support is open Monday to Friday 07:00 - 16:30 CET. Outside Sweden, contact your authorised Vastra service partner first. Questions that the service partner cannot solve are handled by Vastra Technical Support (Level 2). Always state the serial number and the running hours.

## Service record

| Date | Running h | Task(s) | Codes in log | Parts used | Name |
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